Ex.No.12

Examine Network Address Translation (NAT) using Cisco

Packet Tracer

Date:

AIM:

To configure and examine Network Address Translation (NAT) using Cisco Packet Tracer and verify communication between a private network and a public network.

Requirements

- Cisco Packet Tracer
- 1 Router (Cisco 1941)
- 2 Switches (Cisco 2960)
- 2 PCs
- Copper Straight-Through cables

Network Topology

Private Network Public Network

PC0 -------- Switch0 -------- Router -------- Switch1 -------- PC1

192.168.1.10 NAT Router 203.0.113.10

G0/0 Public PC

192.168.1.1

|

G0/1

203.0.113.1

IP Addressing

Device	Interface	IP Address	Subnet Mask
PC0	FastEthernet0	192.168.1.10	255.255.255.0
Router	G0/0	192.168.1.1	255.255.255.0
Router	G0/1	203.0.113.1	255.255.255.0
PC1	FastEthernet0	203.0.113.10	255.255.255.0

PROCEDURE

Step 1: Create the topology

1. Open Cisco Packet Tracer.
2. Place:
   - Two PCs.
   - Two 2960 switches.
   - One 1941 router.
3. Connect:
   - PC0 → Switch0.
   - Switch0 → Router G0/0.
   - Router G0/1 → Switch1.
   - Switch1 → PC1.
4. Use Copper Straight-Through cables.

Step 2: Configure PC0

Go to:

PC0 → Desktop → IP Configuration
Enter:
IP Address : 192.168.1.10
Subnet Mask : 255.255.255.0
Default Gateway : 192.168.1.1

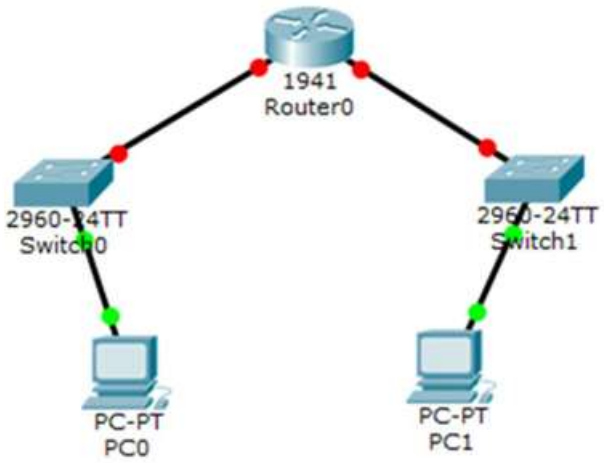
Step 3: Configure PC1
Go to:
PC1 → Desktop → IP Configuration
Enter:
IP Address : 203.0.113.10
Subnet Mask : 255.255.255.0
Default Gateway : 203.0.113.1

Step 4: Configure the Router
Open:
Router → CLI
Configure the inside interface:
Router> enable
Router# configure terminal
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# ip nat inside
Router(config-if)# no shutdown
Router(config-if)# exit
Configure the outside interface:
Router(config)# interface gigabitEthernet 0/1
Router(config-if)# ip address 203.0.113.1 255.255.255.0
Router(config-if)# ip nat outside
Router(config-if)# no shutdown
Router(config-if)# exit

Step 5: Configure NAT
Create an access list to identify the private network:
Router(config)# access-list 1 permit 192.168.1.0 0.0.0.255

Configure PAT (NAT Overload):
Router(config)# ip nat inside source list 1 interface gigabitEthernet 0/1 overload
This allows multiple private IP addresses to share the router's outside IP address.
Step 6: Verify the Router Configuration
Execute:
Router# show ip interface brief

Output:



Result:

Thus, Network Address Translation (NAT) was successfully configured and examined using Cisco Packet Tracer. The private IP address 192.168.1.10 was translated into the public IP address 203.0.113.1, and successful communication with the public network was verified.